

PROJECT-Q Workbook - 10

30-Day Quantum Computing Challenge

DAY - 10

Controlled-NOT (CNOT) Gate

Learning Objectives

By the end of this chapter, you will be able to:

- ✓ Understand the purpose of the Controlled-NOT (CNOT) gate.
 - ✓ Distinguish between the **control qubit** and **target qubit**.
 - ✓ Predict the output of a CNOT operation using truth tables.
 - ✓ Represent the CNOT gate mathematically.
 - ✓ Implement the CNOT gate using Qiskit.
 - ✓ Understand why the CNOT gate is fundamental to quantum computing.
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Quick Recap

Before today's lesson, you learned:

- A multi-qubit system behaves as a single quantum system.
- Two qubits have four computational basis states:
 $|00\rangle, |01\rangle, |10\rangle, |11\rangle$
- The state space grows exponentially as 2^n .
- Multiple qubits can be represented and measured using Qiskit.

Today, we'll learn how **one qubit can control another**, introducing the first operation that acts on **two qubits simultaneously**.

Why Do We Need the CNOT Gate?

So far, every quantum gate you've used (X, Y, Z, H) acted on **one qubit at a time**. However, practical quantum algorithms require qubits to **interact**. The CNOT gate enables this interaction. It allows the state of one qubit to determine whether another qubit should change.

This simple idea is the foundation of:

- Quantum entanglement
- Bell states
- Quantum teleportation
- Error correction
- Grover's algorithm
- Shor's algorithm

What is the CNOT Gate?

The **Controlled-NOT (CNOT)** gate is a **two-qubit quantum gate**.

It consists of:

- **Control Qubit** - Determines whether the operation should occur.
- **Target Qubit** - The qubit that may be flipped.

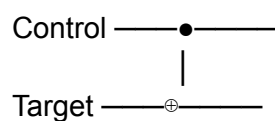
Rule

If the **control qubit is $|1\rangle$** , apply an **X gate** to the target qubit.

If the **control qubit is $|0\rangle$** , do nothing.

This conditional behavior makes the CNOT gate fundamentally different from single-qubit gates.

Visual Representation



- • = Control Qubit
- ⊕ = Target Qubit (NOT operation)

The vertical line indicates that the two qubits are connected through the CNOT operation.

Understanding Through Examples

Example 1

Input:

Control = 0

Target = 0

Since the control qubit is **0**, the target remains unchanged.

Output: 00

Example 2

Input:

Control = 0

Target = 1

Again, the control is **0**.

Output: 01

Example 3

Input:

Control = 1

Target = 0

Now the control qubit is **1**, so the target flips.

Output: 11

Why Is It Called "Controlled-NOT"?

The target qubit undergoes a NOT (X) operation **only when** the control qubit is $|1\rangle$.

Thus, the NOT operation is **controlled** by another qubit.

This conditional logic is why the gate is named **Controlled-NOT**.

Circuit Symbol

A CNOT circuit looks like this:

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$


The diagram shows a CNOT gate circuit. The top qubit, labeled $|a\rangle$ on the left and $|a\rangle$ on the right, is the control. It has a solid black dot. The bottom qubit, labeled $|b\rangle$ on the left and $|a \oplus b\rangle$ on the right, is the target. It has a circle with a plus sign. A vertical line connects the control dot to the target circle.

If q0 is the control and q1 is the target, the target flips only when q0 = $|1\rangle$.

Implementing CNOT in Qiskit

Step 1 - Import Qiskit

```
from qiskit import QuantumCircuit
```

Step 2 - Create Two Qubits

```
qc = QuantumCircuit(2,2)
```

Step 3 - Apply an X Gate

```
qc.x(0)
```

This prepares the control qubit in the state $|1\rangle$.

Step 4 - Apply the CNOT Gate

```
qc.cx(0,1)
```

The syntax is: `qc.cx(control, target)`

Here,

- Control = qubit 0
- Target = qubit 1

Step 5 - Measure Both Qubits

```
qc.measure([0,1],[0,1])
```

Complete Program

```
from qiskit import QuantumCircuit
qc = QuantumCircuit(2,2)
qc.x(0)
qc.cx(0,1)
qc.measure([0,1],[0,1])
print(qc.draw())
```



Expected Output

Initially, $|00\rangle$

After applying, `qc.x(0)`

the state becomes $|10\rangle$

Applying, `qc.cx(0,1)`

changes the target qubit.

Final state: $|11\rangle$

Real-World Importance

The CNOT gate is one of the most frequently used gates in quantum computing because it enables qubits to interact.

It is essential for:

- Creating entangled states
- Quantum teleportation
- Quantum error correction
- Quantum algorithms
- Building larger quantum circuits

Without two-qubit gates like CNOT, quantum computers would be limited to independent qubit operations and could not achieve the computational advantages associated with entanglement.

Common Beginner Mistakes

✗ "The target qubit always flips."

✓ No. The target flips **only if** the control qubit is $|1\rangle$.

✗ "Both qubits change."

✓ The control qubit remains unchanged. Only the target may change.

✗ "CNOT is just another X gate."

✓ No. The X gate acts on a single qubit, whereas the CNOT gate applies an X operation conditionally based on the state of another qubit.

Remember This

- CNOT is a **two-qubit gate**.
 - It consists of a **control qubit** and a **target qubit**.
 - The target flips only when the control qubit is $|1\rangle$.
 - CNOT is represented by a **4×4 matrix**.
 - It is the foundation for entanglement and many quantum algorithms.
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Math Corner

DAY 10 – FORMULA SHEET

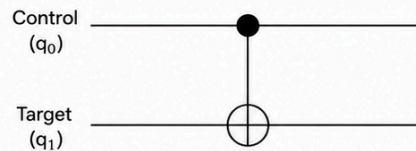
CNOT (CONTROLLED-NOT) GATE

1. CNOT Rule

If the **control** qubit is $|1\rangle$,
apply an X (NOT) gate to the target qubit.
If the **control** qubit is $|0\rangle$, do nothing.

Control qubit $\rightarrow \bullet$
Target qubit $\rightarrow \oplus$ (X operation)

2. Circuit Representation



3. Truth Table (Computational Basis)

Input (Control Target)	Output (Control Target)
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 11\rangle$
$ 11\rangle$	$ 10\rangle$

4. Matrix Representation

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Basis order:
 $|00\rangle, |01\rangle,$
 $|10\rangle, |11\rangle$

5. Action on Basis States

$\text{CNOT } |00\rangle = |00\rangle$
 $\text{CNOT } |01\rangle = |01\rangle$
 $\text{CNOT } |10\rangle = |11\rangle$
 $\text{CNOT } |11\rangle = |10\rangle$



6. General Action

CNOT acts on two qubits as:

$$\text{CNOT } |c, t\rangle = \begin{cases} |c, t\rangle & \text{if } c = |0\rangle \\ |c, t \oplus 1\rangle & \text{if } c = |1\rangle \end{cases}$$

where c = control qubit, t = target qubit

7. Qiskit Implementation

```
from qiskit import QuantumCircuit
qc = QuantumCircuit(2, 2)
qc.cx(0, 1)    # control = 0, target = 1
qc.draw()
```

Syntax: `qc.cx(control, target)`

8. Key Properties

- The control qubit is never changed.
- The target qubit flips only when control = $|1\rangle$.
- CNOT is a reversible (unitary) operation.
- CNOT is its own inverse: $\text{CNOT}^2 = I$
- Essential for entanglement and quantum algorithms.



Remember: CNOT is the first two-qubit gate that creates interaction between qubits.

End-of-Day Summary

Today, you learned:

- The purpose of the CNOT gate.
- The roles of the control and target qubits.
- How CNOT transforms the computational basis states.
- The matrix representation of CNOT.
- How to implement a CNOT gate using Qiskit.
- Why CNOT is indispensable in quantum computing.

Top Verified Learning Resources

1. **IBM Quantum API Documentation - CXGate (Official)**
<https://quantum.cloud.ibm.com/docs/en/api/qiskit/qiskit.circuit.library.CXGate>
2. **Quantum Computing UK - Controlled-NOT Gate Tutorial**
<https://quantumcomputinguk.org/tutorials/controlled-not-gate-program>
3. **The Open University - Controlled NOT Gate**
<https://www.open.edu/openlearn/science-maths-technology/introduction-quantum-computing/content-section-7.3.2>
4. **Computational Mindset - NOT and CNOT Operators**
<https://computationalmindset.com/en/quantum-computing/not-cnot-operators.html>
5. **Recommended Video**
<https://www.youtube.com/watch?v=UyrEqp9eQwQ>
6. **Recommended Video**
<https://www.youtube.com/watch?v=aCOsqL-jlOo>

